

Amrit Bhardwaj

amritbharadwaj4@gmail.com | +91-9508827296 | LinkedIn | GitHub | Portfolio | CP Ratings

Education

Gati Shakti Vishwavidyalaya (Central University)

Vadodara, India

B.Tech in Electronics and Communication Engineering (Specialization in rail engineering)

CGPA: 8.53

July 2027

Technical Skills

Languages: C++, JavaScript (ES6+), TypeScript, Python

Frontend: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS, ShadCN UI

Backend: Node.js, Express.js, REST APIs, WebSockets, WebRTC, JWT Auth, Mongoose, Redis, Pub/Sub, Webhooks, Worker Queues

ML & Data Science: Scikit-learn, XGBoost, CatBoost, Pandas, NumPy

NLP & LLM Tooling: SpaCy, LangChain, LangGraph, RAG, Vector Embeddings, Gemini API

Databases: PostgreSQL, MongoDB, SQLite, Redis (Sorted Sets)

System Design: Event-Driven Architecture, Producer-Consumer Patterns, Leaderboard Archival, Pub/Sub Messaging

Tools & Platforms: Git/GitHub, Docker, Postman, Jupyter, VS Code, AWS S3, EC2, CI-CD PIPELINE

Experience

Frauscher Sensor Technology 

(May 2026 – Present) Mumbai

Project Management Intern

- Built a **Windows background service** (Node.js, TypeScript, strict mode) that monitors shared folders via **chokidar**, detects file-write completion using a polling-based stability check, and uploads validated tender documents to Amazon S3 via resumable multipart uploads *This idea was by me and whole build process was also done by me*.

Mumbai Metropolitan Region Development Authority (MMRDA)-DB RAIL 

(May 2025 – June 2025) Mumbai

Data Science & Machine Learning Intern

- Engineered ML-based solutions to optimize rolling stock scheduling, successfully analyzing metro operational datasets to tune speed parameters and reduce headway latency.

Hibiscus Tech 

(Feb 2026 – Present) Remote

Full Stack Production Engineer (Freelance)

- Built and shipped the real-time leaderboard system using Redis Sorted Sets (ZADD/ZREVRANK), implementing monthly archival snapshots into PostgreSQL to preserve historical rankings at scale.
- Developed the platform's analysis section end-to-end, taking it from design through production deployment with data visualization components and player performance metrics.
- Engineering chess puzzle generation logic using heuristic-based approaches to dynamically serve puzzles matched to player skill.

Projects

DRISHTI:Traffic Intelligence Platform 

FastAPI, Next.js

- Built a full-stack AI triage system for Bengaluru Traffic Police, predicting incident priority (High/Low), estimated clearance time, and officer deployment requirements from free-text citizen complaints in English, Kannada, and Hindi.
- Engineered a **MultiOutputRegressor** (Random Forest, 240 trees) predicting 9 simultaneous resource targets — constables, ASI, SI, inspectors, barricades, tow units, medical units, diversion confidence — from derived impact level, urgency score, and crowd size features; wired to a weekly retraining pipeline consuming resolved incidents from PostgreSQL.

Sosyo – Privacy-First Real-Time Social Networking Platform 

Express.js, PostgreSQL, Redis

- Architected and developed a location-aware social networking platform supporting user discovery, mutual friendships, private messaging, online presence tracking, and nearby active friend recommendations.
- Designed scalable backend services using Node.js, Express.js, PostgreSQL, and Redis, implementing secure JWT-based authentication, OTP verification, session management, and role-based authorization.
- Engineered real-time system architecture for messaging, typing indicators, read receipts, presence tracking, friendship workflows, and Redis-backed caching to support low-latency user interactions and scalable event propagation.

Achievements & Certifications

TCS iON NQT - IT Certification: Scored 89% in Programming (C++) and top percentile in Cognitive Ability.

Competitive Programming: LeetCode Max Rating 1726 (Top 15%), CodeChef (1300+), Codeforces (1100+).

Among Top 1.6k teams Selected for the **Flipkart** Gridathon Round 2

Wabtec Exceed 3.0 Among top 25 teams to qualify for Wabtec Exceed Hackathon 3.0

Awarded as Junior Scientist - Awarded as junior scientist by National Children Science Congress by Department of Science and technology **Part of the KVS National team for the JNMSEE-EXHIBITION**